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Agile design process for additive manufacturing, an exploratory study

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ABSTRACT: This study explores the use of agile methods to support Additive Manufacturing (AM) in transitioning from R&D to production. Using a consumer goods company division as a case study, the research examines how agile methods facilitate flexibility, collaboration, and innovation despite challenges such as the materiality of products methods inconsistencies. Findings reveal how tailored agile practices designed for Additive Manufacturing enhance technology readiness and identify areas for improvement, including stakeholder engagement and role alignment. Recommendations are proposed to refine an Agile Design Process Model for Additive Manufacturing and improve technology maturation.

KEYWORDS: design for additive manufacturing (DfAM), new product development, design process, agile development, agile management

1. Introduction and background

1.1. The need for agility

Organizations are exposed to increased levels of complexity and dynamism due to the pace of change of market landscapes and the rapid introduction of new technologies (Pavlov & Micheli, 2022). As organizations experience fast growth, the organizational structures, the methods and the processes employed, require adaptation (Turetken et al., 2017). Already more than 50 years ago, Moore (1970) illustrated that scaling-up promising results of R&D activities in the production context is challenging due to the "translation gap", consisting of decreasing product lead times, different systems requirements and technological changes. Nowadays, technological complexity is ever increasing which makes the transfer of R&D outcomes to production challenging (Flores Ituarte et al., 2016). In addition, shorter life cycles of technology and innovation require adaptability and efficiency of manufacturing and production technologies and practice (Hofer et al., 2020). Finally, increasingly complex products lead to collaboration challenges as teams from different corporate functions (such as R&D and production) must interact seamlessly for a successful outcome (Un & Asakawa, 2015). The need for agility, as in contrast to plan-driven approaches, led to the development of flexible methods for project management (Schuh et al., 2017). Agile methods have been widely adopted since the start of the century, mainly by the software development industry, as an effort to rapidly respond to changing requirements. Although the word "agile" was already in circulation for decades (Abbas et al., 2008), the concept with its current understanding has obtained wide recognition after the publication of The Agile Manifesto for Software Development by Beedle et al. (2001). The Agile Manifesto highlighted the importance of adaptability to change and human interactions versus following detailed plans. The Agile movement stressed the importance to embrace the inevitability of changing plans. Under the Agile Manifesto are several approaches for software development, such as Scrum, Kanban, and DSDM, that share its fundamental principles. Regarding physical product development, recent research has indicated that agile practices

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can increase the innovation performance of firms and the success of new product development (Meier & Kock, 2024), especially due to faster adaptation to changes in highly complex and dynamic environments (Gonzalez, 2014).
However, the transferability of the agile mindset to corporate functions beyond New Product Development (NPD) into large-scale serial production is posing challenges revolving mainly around the materiality of products, which limits the number of iteration loops and the production of quick deliverables to obtain early stakeholder feedback (Cooper & Sommer, 2018). The challenges related to collaboration are not limited to intra-firm cross-functional interactions, such as handover procedures between different departments, but are also experienced within external product development partnerships. Furthermore, the introduction of Agile principles into the development process requires careful consideration, and is dependent on the activities of the teams and the level of uncertainty in which they operate (Rigby et al., 2020). Firms decide to undergo collaborative innovation partnerships with external partners for various reasons, such as the acquisition of new knowledge (Xie et al., 2020), and the distribution of specific value-creation activities to more specialized partners (Narasimhan & Narayanan, 2013). Stakeholder engagement, e.g., customer participation, can be instrumental in reacting to technological and market changes (Chen & Liu, 2019).
Complex stakeholder engagement and collaboration require a high amount of agility in the form of creativity and flexibility (Chen & Liu, 2019). As knowledge flows vary in intra-firm and inter-firm settings, management approaches must be adapted accordingly (Schleimer & Faems, 2016). Consequently, finding the right approach to efficient stakeholder interactions requires resources (Kaur & Lodhia, 2019). Potential conflicts in working styles and reduced autonomy in decision-making processes are additional challenges that need to be addressed (Andersen & Gadde, 2019).

1.2. Maturity of additive manufacturing for production

Additive Manufacturing (AM), or 3D printing in its more well-known name, has retained the title of a promising and versatile prototyping technology for many years, but not yet optimal for serial production of high surface quality polymer retail products. AM is adopted for all the advantages it provides, such as creating products with complex, changing and custom designs without increasing the manufacturing process complexity (Stavropoulos et al., 2023). Additive Manufacturing is an example of a complex, developing technology that requires custom design supports to reach full maturity. AM, if used as a substitution of traditional methods, has the potential for radically changing the product development process, for which designers need a revised and custom process. This has called for the creation of custom Design for Additive Manufacturing (DfAM), to support new product paradigms (Gibson et al., 2021; Schmitt et al., 2024). According to a recent industry report, Additive Manufacturing solutions are most advanced in the automotive, aerospace, defense, medical, and dental sectors (AMPOWER, 2024). Creating intricate geometries, providing additional functionalities, and reducing manufacturing steps (Blösch-Paidosh & Shea, 2022) are only some of the benefits that push organizations to adopt and invest in AM technologies. Although Additive Manufacturing technologies are spreading to an increasing number of industries, the technical readiness is not high enough yet to support serial production in all contexts. The transferability of technologies to other contexts is limited, among other factors, by the different technical requirements that the final products must meet, such as dimensional accuracy and homogeneity of outputs. In other words, the current technological maturity of AM machines is sufficient for certain industries, but not for others. This makes the industrial use of AM for mass production an uncertain and unstable business, with a challenging road ahead before reaching full maturity. Nevertheless, Additive Manufacturing is still an attractive solution for companies, and its adoption is expected to have a double digit growth in the next 5 years (AMPOWER, 2024). Compared to traditional manufacturing processes, such as injection molding or subtractive manufacturing, AM technologies are still in their formative phase (Ghobadian et al., 2020). However, when comparing the use of AM across different industries, some sectors, such as medical and aerospace, exhibit high maturity, while others, like consumer goods, are still in the early, adolescent, stages of development (Stavropoulos et al., 2023). The maturation of AM technologies from the R&D to the production level represents a complex endeavor, as it requires the interaction of diverse team functions as well as cross-collaboration of firms and industries. The need for collaboration, flexibility and fast adaptation to changes constitutes an opportunity for the use of Agile approaches to develop AM technologies. The potential of Additive Manufacturing, yet not fully exploited, can be harvested through improved design supports (Schmitt et al., 2024).

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As outlined, recent research has emphasized the challenges that need to be overcome to exploit the potential of agile management approaches in the context of R&D and new product development for Additive Manufacturing. Collaboration among internal stakeholders in cross-functional teams, and partnerships with external technology providers and customers, are instrumental in the successful development of complex product portfolio. This research aims to address how agile approaches can be applied through a design process model for Additive Manufacturing, for improved collaboration and product development success. Based on results from an in-depth case study highlighting both successes and learnings, an Agile design process model is proposed including initial recommendations relevant for practice and theory.

1.3. The case of "Design4AM"

This research is conducted at the Additive Manufacturing division of a company that specializes in the manufacturing of high-volume, high-variance durable plastic consumer goods, which we will refer to as "Design4AM". Design4AM carries out research and development activities to mature AM technologies for creating new production lines for mass production, as well as to support existing manufacturing processes. Their work revolves around the discovery, development and maturing of advanced manufacturing technologies. This department represents the ambitions of the firm to intensify, structure and invest in their AM capabilities. Soon after its creation, Agile methods have been introduced to the team, as an experiment to test their efficacy at the company, as well as to support the high ambitions of the project. One crucial moment in the life of the department was the expansion from R&D to production, which caused a fast growth in the workforce. The AM team was initiated with 7 people in 2018 and then grew gradually during the course of 4 years. From 2022 to 2024, the team doubled in just two years reaching a current total of about 50 employees in its 6th year, consisting primarily of AM engineers and their respective managers, a few Agile coaches and additional department support roles. Creating and maintaining collaboration relationships with different types of stakeholders generates an increasing need for flexibility and adaptability to change in the department. Moreover, introducing disruptive production technologies into a traditional setting causes resistance from stakeholders. As previously mentioned, AM technologies lack of availability of mature solutions to support mass production, as well as solid design process models. For these reasons, we analyze the case firm with the aim to understand their current ways of working, the use of Agile for physical product development, and how product development processes can be supported through the design of refined design process models to identify and strengthen influencing criteria for success.

2. Research approach

Following the Design Research Methodology (DRM) (Blessing & Chakrabarti, 2009), we develop a Descriptive Study to set the basis of our research. The Descriptive Study is part of the initial phases of the DRM, and we use it to design a Reference Model that serves as a basis for the design of an Impact Model where to encapsulate recommendations for design improvements. We use a qualitative research approach, and collect and analyze data following an inductive process, as per Gioia (2021). The data is collected through semi-structured interviews, observations, archival documents, and informal conversations. We consider the interviewees as "knowledgeable agents" (Gioia, 2021) who allow in-depth insights into team dynamics, departmental work methods, and their familiarity with these concepts. General questions were posed to avoid leading participants, with probing questions ensuring reported experiences represented genuine challenges. However, secondary data sources were instrumental in identifying bias in responses due to team culture. The interview process has been designed into 8 steps, inspired by Kallio et al. (2016), Moser and Korstjens (2017), and Adeoye-Olatunde et al. (2021). The data collection and analysis protocol is shown in the model of Figure 1. Data analysis is carried out iteratively throughout the process. The sampling technique has followed the purposive sampling approach (Moser & Korstjens, 2017). A total of 14 interviews have been conducted, for an average length of 1 hour each (approx. 100,000 words in total). Approximately 60 hours' worth of fieldnotes have been recorded and complemented by the company's archival documents. The dataset was processed and analyzed using NVivo. Approximately 600 participant-informed codes have been captured during the analysis process. These codes have been grouped under 78 concepts, maintaining the participant words,

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and resulting in the 1st order concepts, as per Gioia (2021). 14 2nd order themes have been identified by further grouping the concepts. Finally, 4 aggregate dimensions emerged as result.
The results of the data collection have been used to design multidimensional models of the approach to product development in use at Design4AM. The learnings deriving from the data have been used to design initial mediating strategies for further successful adoption of agile methods for Additive Manufacturing for mass production.

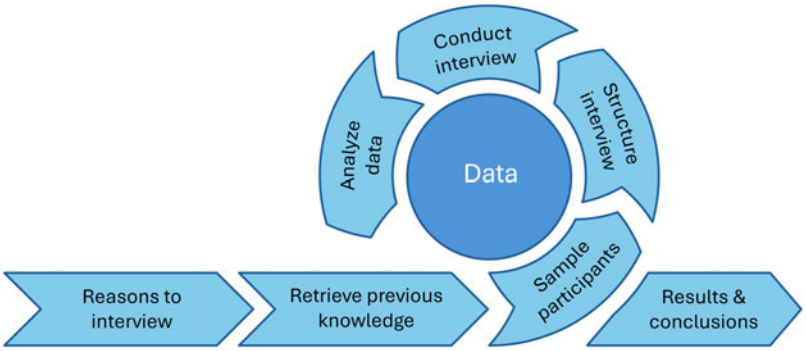


Figure 1. Protocol design for semi-structured interviews

3. Results

3.1. Agile to support AM for serial production: Design4AM

The data collected at the company has been analyzed and used to model the current ways of working of Design4AM, represented in Figure 2. The team has adopted a set of different approaches to manage the development of technologies for AM, but a model to describe their ways of working was not designed yet. With the use of fieldnotes and interviews, the ways of working of Design4AM have been modelled in multiple dimensions. Figure 2 describes organizational structure, decision-making and maturity of activities for product development, complemented by product management approaches in use. The company Design4AM is part of, follows a custom corporate product development (CPD) framework, comparable to the Agile-Stage-Gate model developed by Cooper and Sommer (2016). Figure 2 shows where the activities of Design4AM take place, with respect to the maturity scale of the CPD. The rapid department growth was managed by forming several different product teams, divided into two main areas: Research & Development, and technology maturing for production. The aim is to deliver AM solutions while upholding the research capabilities. The product teams work to enable AM technologies for consumer goods and deliver them to specialized teams that run the operations. The first team is responsible for sourcing technologies and establishing partnerships with suppliers and universities. The middle team improves the technologies for the scope of the company, and the last team in line exploits the technologies as an alternative option to traditional manufacturing. All of the mentioned functions are essential to the success of the team. Firstly, the partnerships with the suppliers are of a symbiotic type, where constant external involvement by the first team ensures fast development at both ends. Secondly, the refinement of the technologies by the second team ensures to push the technological boundaries in the AM industry, and thirdly the adoption of AM technologies in the third team supports the company's needs and extend its product portfolio, which is critical, not only for internal productivity needs, but also to test the consumer goods market for additively manufactured products. To support its activities, the department adopted Agile methods and a scaled Agile framework for managing all activities including design processes within the corporate Stage-Gate model for new product development. The production teams use a version of Scrum (i.e. Agile framework for managing work in short, iterative sprints, focusing on collaboration and continuous improvement, with regular events referred to as "ceremonies"), while the R&D teams use a version of Scrumban (i.e. Agile framework that combines Scrum's structured approach with sprints and Kanban's visual workflow and flexibility). Each team is managed by a Product Owner (PO). The core teams are supported by "satellite teams" who have side functions such as IT and chemical engineering. To scale

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the Agile design processes of the department, all the POs are placed organizationally together in a management team that is referred to as Scrum of Scrums (SoS).

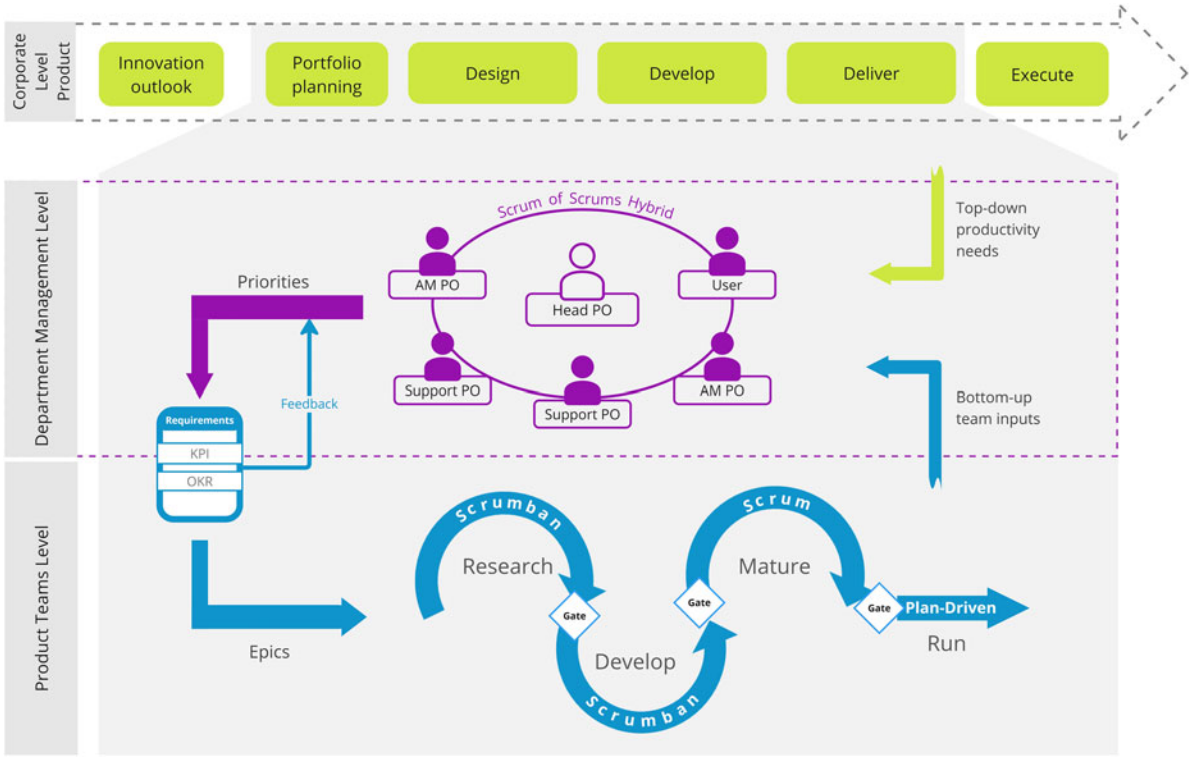


Figure 2. Ways of working model of Design4AM

The SoS sets priorities for the department's activities and includes the core POs and other stakeholders who directly contribute to and support the department's activities. The SoS is also ultimately responsible for go-no-go decisions on the products, represented by the gates in Figure 2. Despite the name, the approach used by this organ borrows only a part of the Scrum structure. The members, in fact, lead teams who use approaches that range from plan-driven to DevOps, including Kanban and Scrum. The SoS is responsible for prioritizing work, based on the productivity needs of the company, but also on the department's own knowledge acquisition. This is done by capturing the yearly goals into Objectives and Key Results (OKR) (i.e. a goal-setting framework used to define and track objectives and their outcomes). These, together with Key Performance Indicators (KPI), are translated into Epics (i.e. Scrum artefacts) that are presented to the product teams, who split them into appropriate chunks of work with the help of an Agile coach. This approach has brought significant results. The stakeholder alignment obtained by grouping all relevant (core and supporting) teams under one organ, the SoS, has allowed Design4AM to produce and launch products with high quality characteristics (e.g. dimensional accuracy, surface roughness, mechanical strength and functionality) not achieved before in AM of consumer goods. This was confirmed by market tests run by the firm, that analyzed the perception of users on additively manufactured products, compared to traditionally manufactured ones. The public reception was positive, as demonstrated by the increase in price registered after launch, as an effect of the high demand. The SoS manages to bridge the gap between traditional pre-existing hierarchical structures that push for productivity needs from the top down, and newly-created Agile product teams that voice their needs from the bottom-up. From the analyses it also emerged that the work of Design4AM couldn't proceed independently from the industry, but that throughout the years it has actively contributed to shaping the industrial landscape and the maturity of AM machines. The approaches that are applied in our reference model have demonstrated a range of benefits from Agile in physical product development. We find that Design4AM experiences similar benefits already described by Sommer et al (2015) in their study. These include, among others, increased design flexibility – proven by the capacity of Design4AM to

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immediately adapt to changes and run frequent design iterations; improved portfolio prioritization, which has been achieved successfully in the SoS thanks to its structure that incorporates core and supporting AM activities under one organ; and improved team morale – as captured not only by direct observations and achieved OKR, but also by the yearly company statistics.

The company's approach has yielded significant advancements in mass production technologies, delivering customer-valued improvements in geometry, manufacturing ease, cost savings, customization, and product functionality. Design4AM has the potential to improve further on its transition from R&D to production. The department aims to advance the maturity of Additive Manufacturing. This ambition necessitates close collaboration with technology manufacturers, and an improved designed for the product development process. The interaction with external suppliers and in-company stakeholders, along with the internal collaborative innovation setup, represents one of the department's most significant challenges.

3.2. Agile for AM: successes, learnings and recommendations

The analysis of the model in use at the department, compared with the data collected through semi-structured interviews of contributors at different functions and hierarchical levels (i.e. engineers, managers and Agile coaches), has provided us with a preliminary overview of the successes and challenges of such an approach to product development management in Additive Manufacturing. These provide relevant insights into practice, while setting the scene for recommendations to improve design processes in the case of Additive Manufacturing for mass production. Design4AM, with its adoption of a hybrid approach to product development, has managed to maintain the flexibility necessary to navigate the volatile and uncertain industry in which it operates while successfully cooperating with teams dedicated to traditional manufacturing that employ plan-driven project management approaches. To successfully mature AM technologies for serial production, Design4AM had to directly contribute to the technological advancement of the solutions offered by its partners, through direct investment and involvement, as its usual for new product development activities. The selection of different approaches (i.e. Scrumban, Scrum or plan-driven) based on the product pipeline enabled the different teams to adopt custom methods and tools to ensure the optimal use of resources, and to work with distinct timeframes to accommodate varying speeds in the product development process, which ensures faster learning loops and in turn quicker design iterations. Design4AM has adopted an approach in which the top-down needs of the company are merged with the bottom-up inputs of the product teams, ensuring resources are used in the most efficient way and aligned to deliver to the goals, shared among engineers and top managers. This enables the department to successfully launch and test multiple products in the market, which were received very positively by the end consumers. The continuous attention to adjusting priorities and goals (through Scrum ceremonies and OKR-dedicated events) has enhanced the flexibility of Design4AM and improved not only its reactivity to changing external inputs, but also the proactivity needed to be at the forefront of technological advancement, which cannot always be achieved through the implementation of traditional methods. Design4AM is the department that fully focuses on Additive Manufacturing at the company, but it is supported by other teams inside the firm whose main tasks are not AM-related, but that do provide full support to the core AM team. In summary, Design4AM has effectively created a collaborative AM environment that goes beyond its organizational boundaries. The creation of such environment, through the inclusion of relevant supporting teams inside the Scrum of Scrums, has guaranteed high alignment levels, and has secured top management support for the program. The use of Scrum and Scrumban provides an immediate solution to the issue of quick hardware development, but it still faces some limitations, such as the ones linked to sprint deliverables, Agile-plan-driven interfaces and the adoption skepticism, as identified by previous studies (Cooper & Sommer, 2018). The learnings from the data are firstly organized into themes and then channeled into four aggregate dimensions, as mentioned previously. Figure 3 shows the themes, the aggregate dimensions, and the corresponding recommendations. These learnings, represented by the aggregate dimensions, are: the familiarity with the methods in use influences the successful adoption of those methods; misalignment in expectations and perception of the hierarchical structure impact the frequency of learning loops; the degree of engagement and alignment with internal stakeholders (such as supporting AM teams) and external suppliers affects the flexibility and the

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workflow of the core functions; the physicality of the product affects, among other, the re-allocation of engineers to different functions.

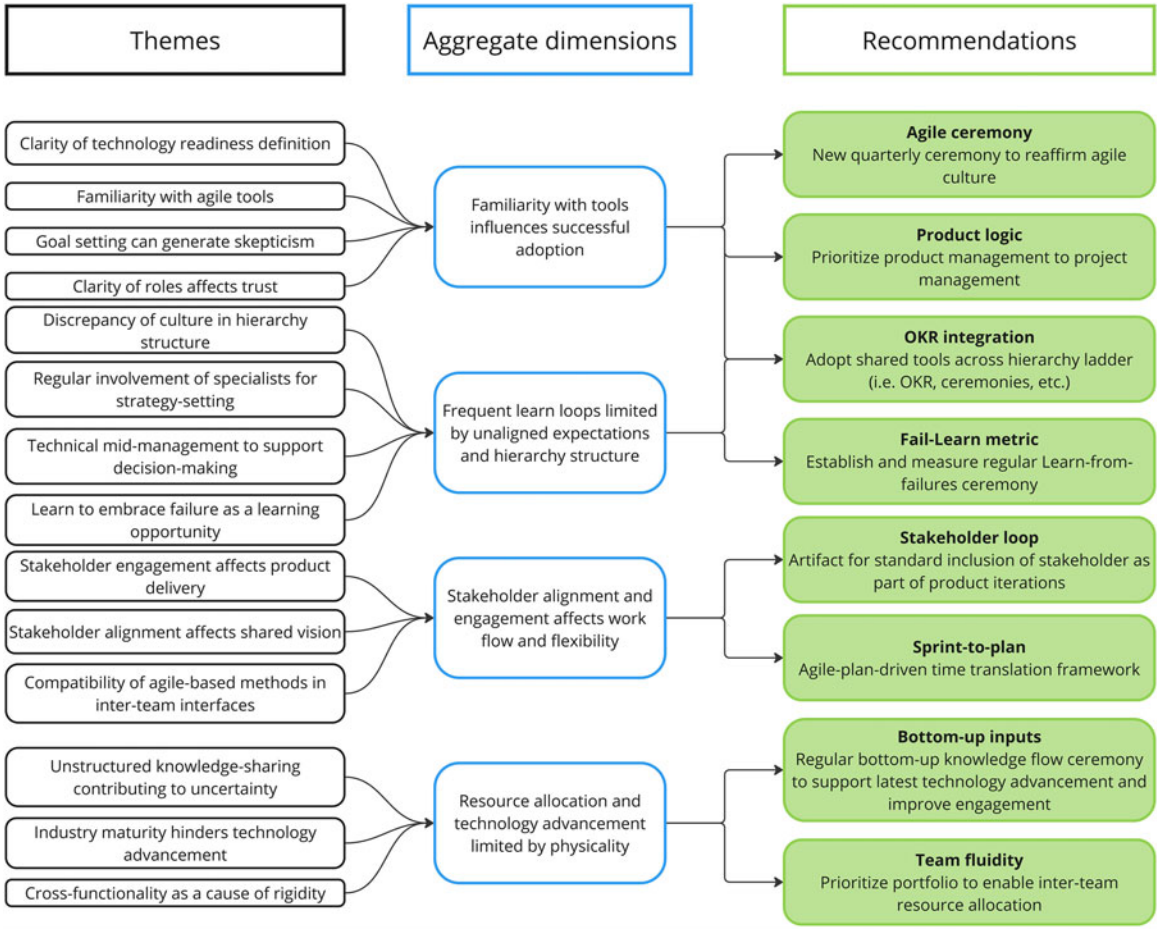


Figure 3. Learnings on using Agile for AM and initial recommendations

In the study we find that the disparity of understandings of Agile culture, models and artifacts can represent a limitation to the successful deployment of such methods, as also discussed in the literature (Cooper & Sommer, 2018; Orejuela et al., 2023). The issue of physicality is also often reported as a central theme around the use of Agile methods for manufacturing (Cooper & Sommer, 2018). The alignment and engagement of contributors and stakeholders is yet another challenge known in the scientific community (Chen & Liu, 2019). While a comprehensive framework represents an opportunity for future research, we propose some recommendations to further improve the use of Agile approaches as a design support for Additive Manufacturing. The recommendations, linked to the aggregate dimensions, include: 1) Agile ceremony: the establishment of a new regular (quarterly, yearly) ceremony dedicated to fostering the agile culture in the team, to improve trust on the models in use; 2) Product logic: the resolution of role conflicts, in terms of Agile and traditional management roles; 3) OKR integration: higher diffusion of OKR across the hierarchy ladder, to solve the familiarity issue and obtaining a more favorable response from contributors; 4) Fail-Learn metric: fostering a culture of "constructive failure" by introducing a new ceremony to actively "collect", celebrate and measure learnings from failures, with the scope of allowing the contributors to change perspective on failures (in terms of design iterations), translating into faster learnings; 5) Stakeholder loop: introducing formal stakeholder involvement phase into the design process to also contribute to faster learnings and more frequent iterations; 6) Sprint-to-plan: establishing a time translation framework to align Agile and plan-driven periodicities; 7) Bottom-up inputs: strengthening the knowledge flow to support decision-making, by introducing a new ceremony to enable the management organ to acquire teams inputs faster; 8) Team fluidity: support re-allocation of contributors through portfolio prioritization.

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These practical recommendations are visualized in Figure 4, in relation to the hierarchical level where they can find application.

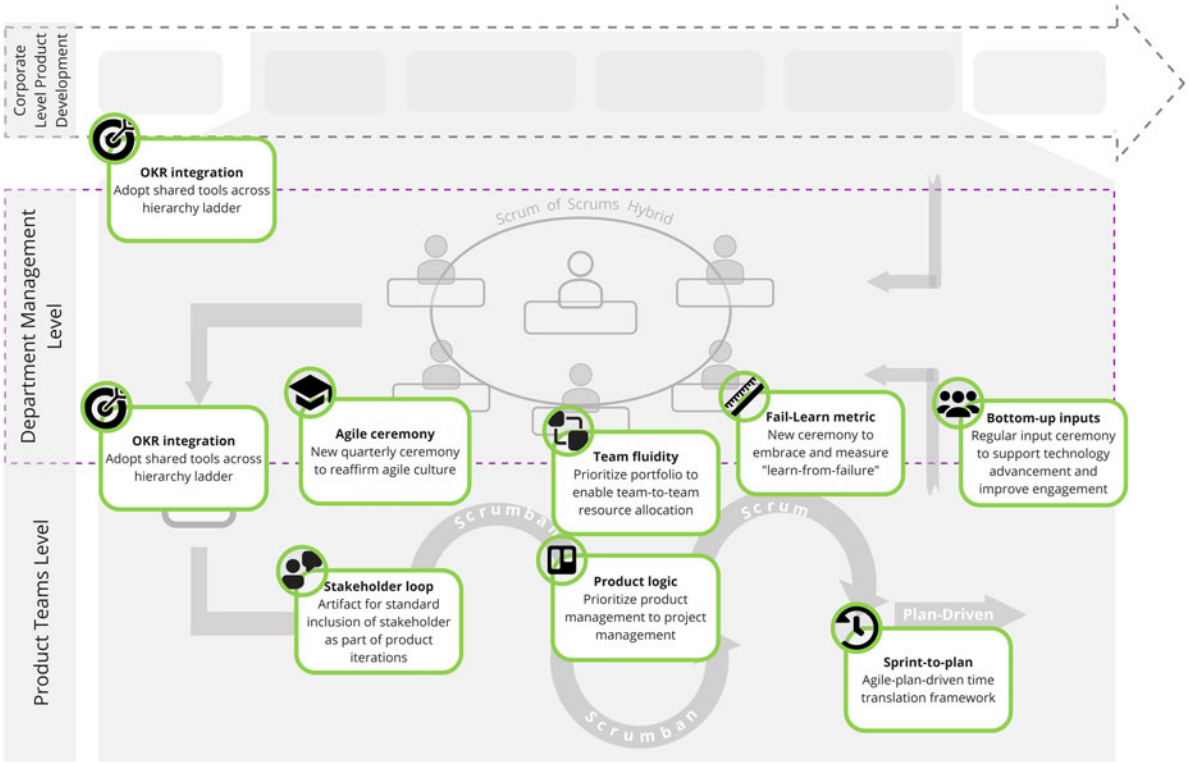


Figure 4. Practical recommendations to support the use of Agile for additive manufacturing

4. Discussion and conclusions

Design4AM represents a leading effort in advancing Additive Manufacturing for consumer goods. Despite notable successes, refining the design process is needed to further support the maturation of AM technologies and enable mass production at scale. Some recommendations outlined in this study, such as supporting teams with tailored Agile artifacts and the use of OKR to bridge the gap between traditional and Agile teams, are already under experimentation. Key strategies include stakeholder inclusion in departmental activities and educational initiatives to enhance knowledge of AM solutions and increase their appeal among decision-makers. The maturation of technologies with the potential to disrupt products, processes and paradigms, such as AM technologies, faces challenges at multiple levels. In this study we uncovered part of them, which revolve mainly around the collaborative nature of the product development at Design4AM when using Agile methods. Supporting a stronger internal design process can translate into stronger partnerships with stakeholders and sustain the technology advancement itself. The disparity of understanding of agile culture, models and artifacts, the physicality of products and the alignment and engagement of stakeholders, are some of the hurdles to overcome to successfully mature AM technologies for serial production, and are comparable to what discussed in the literature. Therefore, some of the fundamental challenges experienced by the case firm are experienced in other manufacturing contexts. For this reason, the recommendations proposed in this study can find applicability in similar contexts. Agile methods find increasing importance in manufacturing and represent a promising support for the context of Additive Manufacturing, where rapid prototyping and flexibility are paramount. The integration of Agile principles as design support for Additive Manufacturing allows for a more responsive and iterative design process, which is essential for meeting the dynamic demands of today's markets. Agile methodologies facilitate the swift adaptation of designs in response to customer feedback and market changes, nevertheless, their practical application is not easy and requires structures in place to account for the diverse work approaches in use by collaborating teams. This is not yet fully achieved, but the development of context-relevant agile design process model for AM can provide a solution to support the maturation of AM technologies beyond the current technical limitations. This is an exploratory study, and it is used as a basis for further investigation and the development of an Impact Model of an agile product development process for Additive Manufacturing. The dataset is collected at one firm only,

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however due to its affinity with other physical product development processes, the findings are relevant for broader consideration, especially in contexts of Additive Manufacturing Design Processes. Future work should investigate the implementation of the proposed Agile Design Process Model for Additive Manufacturing and delve deeper into the application of the identified improvement areas.

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